

Industrial Spares Inventory Optimization & Working-Capital Analysis

SAIL Project 9 — Portfolio Report

Tools: Microsoft Excel • Power BI • DAX **Dataset:** Synthetic industrial spares dataset

Executive Summary

This project develops an inventory decision-support model for industrial spare parts. It combines inventory valuation, ABC classification, movement classification, criticality, stock-out risk, supplier lead time and excess-inventory analysis to identify where working capital is tied up and where management attention should be focused.

Metric	Result
Total inventory value	■7.57 Cr
Slow-moving inventory	■2.85 Cr (37.7%)
Non-moving inventory	■93.88 L (12.4%)
Actionable excess inventory	■2.86 Cr
Stock-out risk inventory	■59.61 K
De-procurement review	■79.81 L
Scenario-based potential release	■91.30 L (12.06%)

Headline finding: The model identifies ■2.86 Cr of actionable excess inventory under a definition restricted to non-critical, slow-moving or non-moving materials. A separate conservative scenario estimates ■91.30 L of potential working-capital release. Both require operational validation.

Data note: The dataset is synthetic and was created for portfolio/academic modelling; it does not represent confidential SAIL inventory records.

1. Business Objective

Build a practical inventory-optimization framework to identify slow/non-moving inventory, assess stock-out exposure, distinguish critical spares from potential excess, and estimate the working-capital opportunity associated with inventory rationalization.

2. Methodology

Stage	Analysis	Purpose
Data preparation	Structured 500-material synthetic dataset	Create analytical base
Inventory valuation	Closing Stock × Unit Cost	Measure capital tied up
ABC classification	Cumulative inventory-value analysis	Prioritize high-value materials
Movement analysis	Active / Slow-Moving / Non-Moving	Identify weak movement
Criticality	Critical / High / Medium / Low	Protect important spares
Stock-out risk	Consumption, lead time, safety stock, reorder point	Flag supply risk
Supplier analysis	Inventory value, lead time, stock-out events	Assess procurement exposure
Excess analysis	Stock above calculated reorder point	Estimate excess exposure
Action model	Movement + criticality based recommendations	Convert analysis into actions
BI reporting	Interactive Power BI pages and slicers	Support management decisions

3. Decision Logic

Protect: Critical spares, especially where stock risk exists.

Optimize: High-value slow-moving materials through safety-stock and consumption review.

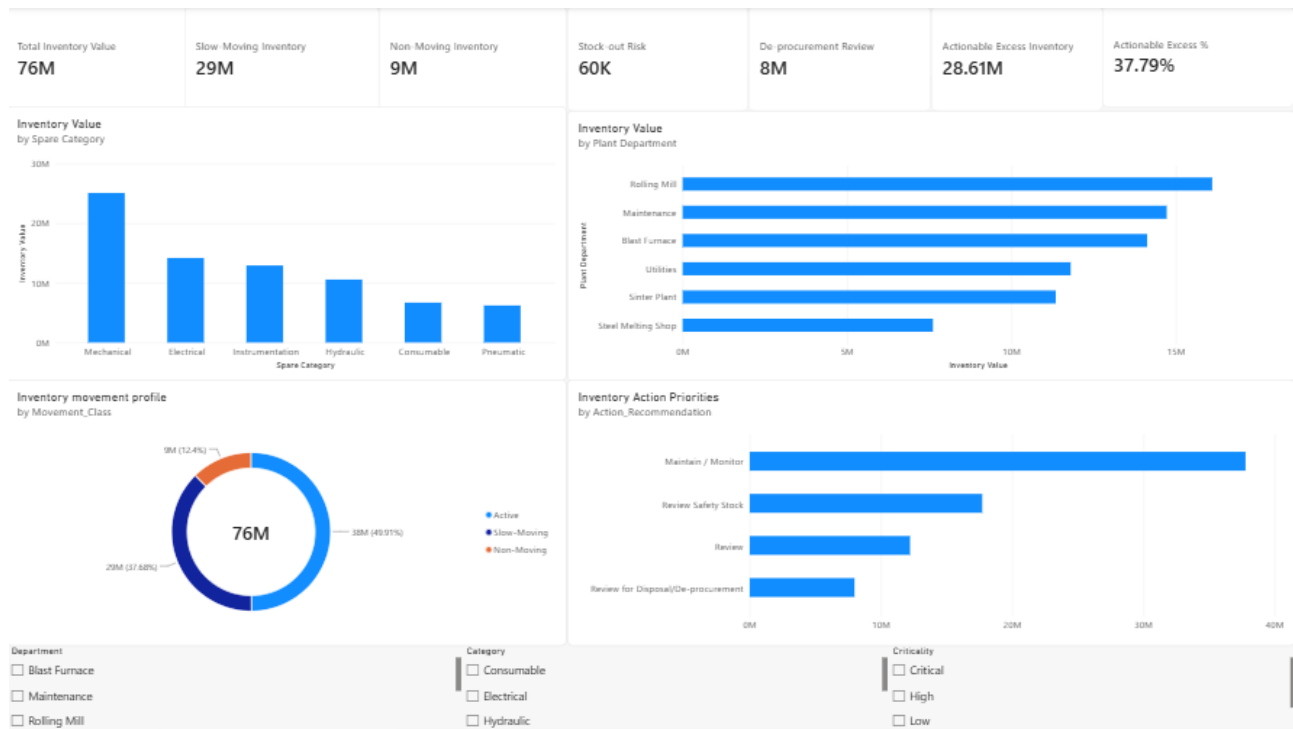
Rationalize: Non-critical slow/non-moving excess for de-procurement, disposal or other disposition review.

Maintain/Monitor: Active inventory with adequate stock.

4. Key Analytical Outputs

The model produced ■5.66 Cr of broad reviewable excess exposure above the calculated reorder-point threshold. A narrower ■2.86 Cr actionable-excess measure excludes Critical materials and focuses on Slow-Moving or Non-Moving inventory. The separate ■91.30 L potential release is a conservative scenario using assumed 70% reduction for disposal/de-procurement review and 20% reduction for safety-stock review; it is not a realized saving.

5. Power BI — Executive Overview



The Executive Overview consolidates inventory value, slow/non-moving exposure, stock-out risk, action priorities and department/category distribution. Slicers allow filtering by department, category and criticality.

6. Power BI — Inventory Optimization & Risk Assessment



The second page provides material-level visibility and procurement context through the action table, potential release and excess measures, criticality/movement matrix, supplier inventory exposure, supplier lead time and stock-out events.

7. Management Recommendations

- 1. Prioritize non-moving and high-value slow-moving inventory.** Investigate high-value items for de-procurement, supplier return where feasible, inter-department transfer or controlled disposal of genuinely obsolete items.
- 2. Reassess safety-stock levels.** Review consumption, lead time, criticality and service requirements before reducing buffers.
- 3. Protect critical spares.** Low movement alone should not trigger disposal; criticality must remain a constraint on inventory reduction.
- 4. Use supplier risk signals.** Review high inventory value together with long lead time and repeated stock-out events to prioritize procurement improvement.
- 5. Focus management effort using ABC classification.** Give high-value A-class materials tighter review and control than low-value items.

8. Limitations & Assumptions

- Dataset is synthetic and designed to demonstrate the requested analytical workflow.
- Reorder-point and safety-stock calculations are simplified portfolio-model assumptions rather than plant-specific policies.
- Potential working-capital release is scenario-based and is not a realized saving.
- Actual implementation requires validation against production/maintenance criticality, approved stocking policies, supplier contracts, procurement constraints and historical demand variability.

9. Project Outcome

The project demonstrates an end-to-end analytics workflow: data preparation in Excel, inventory segmentation and risk logic, scenario-based working-capital analysis, DAX/Power BI reporting, and conversion of quantitative findings into actionable inventory and procurement recommendations.